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RESEARCH ARTICLE

Multi-Objective Approach for Distribution System Planning Considering Stochastic Customer-Owned Distributed Energy Resources

RENATO S. F. FERRAZ¹, RAFAEL S. F. FERRAZ², VANIO F. S. JÚNIOR³,
AND AUGUSTO C. RUEDA-MEDINA¹

¹Department of Electrical Engineering, Federal University of Espírito Santo, Vitória 29075-910, Brazil

²Department of Renewable Energy Technology, Federal University of Piauí, Teresina 64049-550, Brazil

³Department of Engineering and Computing, State University of Santa Cruz, Ilhéus 45662-900, Brazil

Corresponding author: Renato S. F. Ferraz (renatofferraz@gmail.com)

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ABSTRACT The installation of distributed energy resources (DERs) by end-customers has increased significantly in recent years. While customer-owned DERs offer various benefits, their integration adds complexity to distribution system planning due to the unpredictability of their quantity, sizes, and locations. Consequently, they are referred to as stochastic customer-owned DERs in this study. In the specialized literature, optimization-based DER allocation is often explored; however, these allocation problems typically assume that the distribution system operator has control over the allocation of DERs within the grid, whereas real-world investments are predominantly driven by end-customers. Therefore, in this article, a multi-objective planning approach is proposed considering stochastic customer-owned DERs. This approach simultaneously addresses dynamic network reconfiguration, capacitors allocation, and dynamic adjustment of on-load tap changer transformer. The main objective is to minimize power loss costs and improve the system's voltage profile by minimizing a novel voltage consistency indicator (VCI) introduced in this article. The optimization problem is solved using the Multi-objective Cuckoo Search, with the final solution selected through the Fuzzy Decision-making Method. Additionally, the Monte Carlo Method is employed to address uncertainties related to load, DER generation, and the quantity, sizes, and locations of customer-owned DERs. Finally, the results highlighted the importance of conducting a multi-period analysis, as it led to improvements compared to the static analysis. The proposed approach effectively reduced the VCI by 72.41% and minimized the power loss costs by 68.56% compared to the original system.

INDEX TERMS Capacitor allocation, distributed energy resources, multi-objective optimization, network reconfiguration, on-load tap changer transformer, voltage consistency indicator.

I. INTRODUCTION

The increasing deployment of inverter-based distributed energy resources (DERs) in distribution systems has become a prominent trend in recent years. As highlighted in [1], a significant portion of these DERs is integrated into low-voltage networks by end-customers. Nevertheless, their presence has noticeable impacts on medium-voltage systems,

raising challenges related to operational efficiency and grid stability [2]. Consequently, effective planning and optimization strategies are essential to ensure the safe and reliable integration of DERs into modern distribution networks [3]. In addition, the unpredictable nature of the quantity, sizes, and locations of customer-owned DERs—referred to as stochastic customer-owned DERs in this article—adds significant complexity to medium-voltage system planning, requiring robust approaches to address their inherent uncertainty and variability.

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In this context, several strategies are widely adopted in modern distribution networks to ensure efficiency and reliability, even in the presence of DERs, including the installation of on-load tap changer (OLTC) transformers, voltage regulators, and capacitors, as well as network reconfiguration [4]. These strategies can be approached as either static or dynamic, each offering distinct advantages. Static methods rely on fixed decision variables, ensuring simplicity and straightforward implementation. On the other hand, dynamic approaches adjust these variables over time to align with the system's operational conditions, which is particularly advantageous given the inherent variability in demand and DER generation throughout the day [5]. However, this adaptability increases computational complexity, raising the need to assess whether the benefits of dynamic strategies justify their added intricacy compared to static methods.

Therefore, reviewing studies on DER integration, with a focus on planning and operational methods to enhance reliability and manage uncertainties, offers valuable insights into existing solutions and identifies opportunities for advancing planning techniques.

A. RELATED WORK

In the literature, many articles, e.g., [6], [7], [8], and [9], leverage metaheuristic algorithms to determine the optimized allocation and sizing of DERs in distribution systems, aiming to enhance the system's performance. These studies typically focus on objectives like minimizing power losses, reducing voltage deviations, and improving voltage stability. By incorporating operational and physical constraints of both the grid and DERs, these approaches enable secure and efficient DER integration, effectively addressing the challenges posed by high penetration levels and mitigating potential adverse impacts on the distribution network. Despite these benefits, DERs allocation problems assume that the distribution system operator (DSO) can allocate and size DERs within the grid, whereas real-world investments are primarily driven by private initiatives and end-customers.

Additionally, network reconfiguration has emerged as a viable operational strategy for enhancing the performance of distribution systems. Initially proposed in [10] to reduce power losses during normal operating conditions, its applications have expanded its scope to include objectives such as enhancing power quality, increasing voltage security margins, balancing loads, improving reliability, restoring services, and isolating faults effectively, as discussed in [11]. For example, the authors in [12] employ the Harris Hawks Optimization algorithm to minimize power losses and improve the voltage profile through network reconfiguration. Similarly, [13] introduces a novel optimization approach that leverages dynamic network reconfiguration to minimize power losses, voltage deviation, and load imbalance, while addressing the variability and uncertainties inherent in loads and DER

generation. In [14], Particle Swarm Optimization is employed to reduce the overall costs by integrating DER allocation and dynamic network reconfiguration, with a specific focus on uncertainties related to wind speed, irradiance, and load consumption.

In [15], static network reconfiguration is combined with the optimized operation of switchable capacitors and OLTC transformer to minimize power losses in distribution systems, considering smart photovoltaic systems installed at fixed locations. The Multi-agent Deep Deterministic Policy Gradient algorithm is employed in [16] to manage the operation of OLTC transformers and capacitor banks, as well as regulating the reactive power of photovoltaic systems. This study accounts for load and photovoltaic generation uncertainties, with the primary goal of minimizing power losses and mitigating fast voltage fluctuations. The authors in [17] combine the optimized operation of OLTC transformers and capacitors with dynamic network reconfiguration to minimize costs and voltage deviations. The study considers uncertainties in DER generation and load profiles, assuming that OLTC transformers, capacitors, and DERs are installed at fixed positions within the distribution network.

Furthermore, [18] combines network reconfiguration with the allocation of capacitors and DERs to minimize power losses and costs, addressing the dynamic nature of load variations. In [19], a voltage control scheme is proposed to coordinate the OLTC transformer, capacitor banks, and DERs, assuming these devices are installed at fixed locations within the distribution system. This approach improves the voltage profile by defining the OLTC tap positions, capacitor operating modes, and DER output power settings. In [20], a coordinated strategy combining switchable capacitor banks with dynamic network reconfiguration is proposed. The authors highlight that DERs owned by customers are often not optimally located, as their installation is driven by customer preferences rather than system needs. Despite that, the study assumes that these DERs locations are fixed and predefined. This assumption, however, may diverge from real-world conditions, since the placement and capacity of customer-installed DERs are inherently unpredictable, making it challenging to account for them in operational strategies.

In [21], dynamic network reconfiguration is integrated with the allocation of electric vehicle charging stations, switchable capacitors, and DERs to minimize overall costs, voltage deviation, and power losses, while ensuring compliance with network constraints. Meanwhile, [22] employs the Non-dominated Sorting Genetic Algorithm II to maximize DER hosting capacity and minimize power losses through the optimization of volt/var control curves and static network reconfiguration. Notably, this study incorporates stochastic customer-owned DERs, reflecting the uncertainties inherent in their operation and placement. A two-stage optimization strategy is proposed in [23] combining network reconfiguration with the operation of switchable capacitors, OLTC

TABLE 1. Comparison of the proposed approach with other articles.

Article	Optimization approach	Capacitors allocation	OLTC adjustment	Customer-owned DERs	Network reconfiguration	Single-period	Multi-period	Uncertainty	
								Load	DER
[6]	SO	✓	×	×	×	✓	×	×	×
[7]	SO	×	×	×	✓	×	✓	×	×
[8]	MO	×	×	×	×	×	✓	✓	✓
[9]	SO	×	✓	×	×	✓	×	×	×
[12]	SO	×	×	×	✓	✓	×	×	×
[13]	MO	×	×	×	✓	×	✓	✓	✓
[14]	SO	×	×	×	✓	×	✓	✓	✓
[15]	SO	×	✓	×	✓	✓	×	×	×
[16]	SO	×	✓	×	×	×	✓	✓	✓
[17]	SO	×	✓	×	✓	×	✓	✓	✓
[18]	SO	✓	×	×	✓	✓	×	✓	×
[19]	SO	×	✓	×	✓	×	✓	×	✓
[20]	SO	✓	×	✓	✓	×	✓	✓	✓
[21]	MO	✓	×	×	✓	×	✓	✓	✓
[22]	MO	×	×	✓	✓	×	✓	✓	✓
[23]	MO	×	✓	×	✓	×	✓	×	✓
[24]	SO	×	✓	✓	×	×	✓	×	×
This article	MO	✓	✓	✓	✓	✓	✓	✓	✓

✓: Yes. ×: No. MO: Multi-objective. SO: Single-objective.

transformer, and static reactive power compensator. The study considers the presence of DERs; nevertheless, their locations are assumed to be fixed and predefined. In [24], the optimization focuses on the tap position of the OLTC transformer, the tap position of capacitors, and the reactive power of customer-owned DERs, which are also assumed to be installed at fixed and predefined locations.

Table 1 summarizes the main characteristics of the articles discussed in this section. It highlights the optimization approach employed, distinguishing between multi-objective and single-objective strategies. The table also provides details on capacitors allocation, OLTC transformer adjustments, consideration of customer-owned DERs, network reconfiguration, evaluation of single-period and multi-period approaches, and the inclusion of uncertainties in the problem formulation.

Based on Table 1, references [6], [7], [9], [12], [14], [15], [16], [17], [18], [19], [20], [24] adopted single-objective optimization methods, despite the inherent multi-objective nature and trade-offs prevalent in the challenges addressed in this field. Furthermore, only references [20], [22], [24] considered customer-owned DERs, which adds complexity to the distribution system planning problem since it is not possible to optimally allocate and size DERs. Among these studies, only [22] accounted for stochastic customer-owned DERs, where the quantity, sizes, and locations of DERs installed by end-customers are inherently unpredictable. Some of the mentioned articles proposed single-period or multi-period approaches; nevertheless, none of them compared both approaches to determine whether the additional complexity of the multi-period approach is justified compared to the single-period approach. Regarding uncertainties, it is noted that some articles do not account for the uncertainties in loads or DER generation, even though considering these uncertainties is essential to ensure the reliability and robustness of the proposed methodology. Finally, to the best of the authors'

knowledge, the combination of all the aspects outlined in Table 1 has not yet been addressed in the literature.

B. CONTRIBUTIONS

This article presents a novel multi-objective planning approach that simultaneously addresses dynamic network reconfiguration, capacitors allocation, and dynamic adjustment of the OLTC transformer, considering the stochastic nature of customer-owned DERs. In contrast to conventional methods, which often assume deterministic or centrally planned DER deployment, the proposed approach incorporates the uncertainties associated with the number, sizes, and locations of customer-owned DERs. The primary objective is to minimize costs related to power losses and improve the system's voltage profile by minimizing a novel voltage consistency indicator (VCI) introduced in this article. Both single-period and multi-period analyses are conducted to assess the benefits of a dynamic strategy over static methodologies.

The main contributions of this article are:

- 1) Introduction of the VCI, a novel metric for voltage regulation assessment. Unlike conventional voltage deviation indicators, the VCI captures not only the average deviation from the nominal voltage but also the distribution boundaries, excluding outliers. This ensures that the metric remains sensitive to both average and extreme deviations, leading to a more reliable evaluation of the system's performance.
- 2) The proposed approach considers stochastic customer-owned DERs, where the quantity, sizes, and locations are unpredictable variables. In the formulated problem, these variables are randomly generated using a stochastic method. By reflecting real-world scenarios in which DERs are installed by end-customers rather than being strategically allocated by the DSO, the approach offers

a more practical and applicable framework for planning and operating modern distribution networks.

- 3) The proposed multi-objective planning approach simultaneously considers the dynamic reconfiguration of the network, the allocation and sizing of capacitors, and the dynamic adjustment of the OLTC transformer at the substation. This comprehensive approach contrasts with conventional methodologies that optimize these elements separately, improving the overall system's efficiency and coordination.
- 4) A comparative analysis of static and dynamic strategies is carried out to assess their relative performance. The results demonstrate that dynamic approaches improve voltage regulation and reduce power losses, providing quantitative evidence that justifies their practical implementation despite their added complexity.

By integrating these novel elements, this work enhances multi-objective planning strategies for modern distribution networks, offering a framework that is not only theoretically sound but also aligned with real-world operational challenges.

C. ORGANIZATION

This article is structured in 5 sections. Section I presents the introduction, the related work, and the research contributions. Section II outlines the mathematical formulation of the multi-objective optimization problem. The proposed methodology is detailed in Section III. The results of the performed simulations and the corresponding discussions are presented in Section IV. Finally, concluding remarks are provided in Section V.

II. MATHEMATICAL FORMULATION

In this section, the mathematical formulation of the multi-objective optimization problem is presented, encompassing the dynamic reconfiguration of the network, the allocation and sizing of capacitors, and the dynamic adjustment of the OLTC transformer at the substation. The objective functions of this optimization problem are

$$\text{minimize } f_1 = \sum_{h \in \Omega_h} \sum_{ij \in \Omega_l} s_{h,ij} \cdot ec^s \cdot P_{h,ij}^{\text{loss}}, \quad (1)$$

$$\text{minimize } f_2 = \text{VCI}, \quad (2)$$

where

$$\text{VCI} = \sqrt{\frac{(|\sigma_1 - V^n| + |\sigma_2 - V^n|) \cdot \sum_{h \in \Omega_h} \sum_{i \in \Omega_b} |V_{h,i} - V^n|}{|\Omega_h| \cdot |\Omega_b|}}, \quad (3)$$

$$\sigma_1 = \min \{V_{h,i}^{\max}, (\text{quartile}(V_{h,i}, 0.75) + 1.5 \cdot \text{IQR})\}, \quad (4)$$

$$\sigma_2 = \max \{V_{h,i}^{\min}, (\text{quartile}(V_{h,i}, 0.25) - 1.5 \cdot \text{IQR})\}, \quad (5)$$

$$\text{IQR} = \text{quartile}(V_{h,i}, 0.75) - \text{quartile}(V_{h,i}, 0.25). \quad (6)$$

Additionally, the optimization problem is subject to constraints (7)–(21).

$$I_{h,ij} \leq \bar{I}_{ij} \quad \forall h \in \Omega_h, \forall ij \in \Omega_l, \quad (7)$$

$$\underline{V}_i \leq V_{h,i} \leq \bar{V}_i \quad \forall h \in \Omega_h, \forall i \in \Omega_b, \quad (8)$$

$$\sum_{ki \in \Omega_l} P_{h,ki} - \sum_{ij \in \Omega_l} (P_{h,ij} + R_{ij} \cdot I_{h,ij}^2) + P_{h,i}^s = P_{h,i}^d, \quad \forall h \in \Omega_h, \forall i \in \Omega_b, \quad (9)$$

$$\sum_{ki \in \Omega_l} Q_{h,ki} - \sum_{ij \in \Omega_l} (Q_{h,ij} + X_{ij} \cdot I_{h,ij}^2) + Q_{h,i}^s = Q_{h,i}^d, \quad \forall h \in \Omega_h, \forall i \in \Omega_b, \quad (10)$$

$$V_{h,i}^2 - 2 \cdot (R_{ij} \cdot P_{h,ij} + X_{ij} \cdot Q_{h,ij}) - Z_{ij}^2 \cdot I_{h,ij}^2 - V_{h,j}^2 = 0, \quad \forall h \in \Omega_h, \forall i \in \Omega_b, \forall j \in \Omega_b, \forall ij \in \Omega_l, \quad (11)$$

$$V_{h,j}^2 \cdot I_{h,ij}^2 = P_{h,ij}^2 + Q_{h,ij}^2 \quad \forall h \in \Omega_h, \forall j \in \Omega_b, \forall ij \in \Omega_l, \quad (12)$$

$$\text{tap} \leq \text{tap}^{\text{OLTC}} \leq \bar{\text{tap}}, \quad (13)$$

$$V_h^s = V^n \cdot (1 + \text{tap}^{\text{step}} \cdot \text{tap}^{\text{OLTC}}) \quad \forall h \in \Omega_h, \quad (14)$$

$$(P_{h,c}^{\text{DER}})^2 + (Q_{h,c}^{\text{DER}})^2 \leq (S_c^{\text{rated}})^2 \quad \forall h \in \Omega_h, \forall c \in \Omega_c, \quad (15)$$

$$P_{h,c}^{\text{DER}} \geq 0.05 \cdot P_c^{\text{rated}} \quad \forall h \in \Omega_h, \forall c \in \Omega_c, \quad (16)$$

$$-0.44 \cdot S_c^{\text{rated}} \leq Q_{h,c}^{\text{DER}} \leq 0.44 \cdot S_c^{\text{rated}} \quad \forall h \in \Omega_h, \forall c \in \Omega_c, \quad (17)$$

$$-2.2 \cdot \frac{P_{h,c}^{\text{DER}} \cdot S_c^{\text{rated}}}{P_c^{\text{rated}}} \leq Q_{h,c}^{\text{DER}} \leq 2.2 \cdot \frac{P_{h,c}^{\text{DER}} \cdot S_c^{\text{rated}}}{P_c^{\text{rated}}} \quad \forall h \in \Omega_h, \forall c \in \Omega_c, \quad (18)$$

$$\sum_{i \in \Omega_b} (w_i \cdot C^{\text{CB}}) + z \cdot C^{\text{OLTC}} \leq C^{\text{max}}, \quad (19)$$

$$\sum_{ij \in \Omega_l} s_{h,ij} = |\Omega_b| - 1 \quad \forall h \in \Omega_h, \quad (20)$$

$$\sum_{ij \in \Omega_l} |s_{h,ij} - s_{h-1,ij}| \leq \Delta s^{\text{dyn}} \quad \forall h \in \Omega_h. \quad (21)$$

In the mathematical formulation presented before, the sets of time intervals, branches, buses, and customers with DER are represented by Ω_h , Ω_l , Ω_b , and Ω_c , respectively. Accordingly, h , ij , i , and c correspond to the evaluated time, branch, bus, and customer with DER, respectively. The total number of buses in the network is represented by $|\Omega_b|$, while $|\Omega_h|$ denotes the total time horizon considered for the optimization process. Additionally, ec^s is the energy cost associated with the substation; C^{OLTC} and C^{CB} are the investment costs for the OLTC and the capacitors, respectively; and C^{max} is the total available investment budget. Furthermore, Z_{ij} , R_{ij} , and X_{ij} represent the impedance, resistance, and reactance of branch ij , respectively. The rated active and apparent power of the DER for a customer c are denoted as P_c^{rated} and S_c^{rated} ; while $P_{h,c}^{\text{DER}}$ and $Q_{h,c}^{\text{DER}}$ are the active and reactive power generated by each DER at time h , respectively.

The variables $P_{h,i}^s$ and $Q_{h,i}^s$ denote the active and reactive power supplied by the substation at time h and bus i ; $P_{h,i}^d$ and $Q_{h,i}^d$ are the active and reactive power demands at time h and bus i ; $P_{h,ki}$ and $Q_{h,ki}$ are the active and reactive power flows at time h and branch ki ; and $P_{h,ij}$ and $Q_{h,ij}$ are the active and reactive power flows at time h and branch ij , respectively. The active power losses in branch ij at time h is given by $P_{h,ij}^{\text{loss}}$. The variables $\underline{V}_i$ and $\overline{V}_i$ represent the lower and upper voltage limits at bus i , respectively, while V^n denotes the nominal voltage of the distribution system. $V_{h,i}$ and $V_{h,j}$ are the voltage values at time h for buses i and j , respectively, and V_h^s refers to the substation voltage at time h . The interquartile range (IQR) represents the distance between the 75th percentile, quartile($V_{h,i}$, 0.75), and the 25th percentile, quartile($V_{h,i}$, 0.25), of the voltage dataset. σ_1 represents the minimum between the largest observation in the dataset and the 75th percentile plus 1.5 times the interquartile range. Similarly, σ_2 is defined as the maximum between the smallest observation in the dataset and the 25th percentile minus 1.5 times the interquartile range.

The measured and maximum allowable current flow in branch ij are depicted as $I_{h,ij}$ and $\overline{I}_{ij}$, respectively. Moreover, $s_{h,ij}$ is a binary variable indicating the switch status, where 1 represents a closed switch and 0 represents an opened switch; while Δs^{dyn} is the maximum number of switching actions. The variables tap^{OLTC} and tap^{step} represent the tap position and the step size of the OLTC transformer; and $\underline{\text{tap}}$ and $\overline{\text{tap}}$ define the minimum and maximum allowable tap positions of the OLTC transformer, respectively. Finally, z and w_i are binary variables indicating the presence of OLTC transformer and capacitor in the distribution system, respectively.

The first objective function, shown in (1), focuses on minimizing active power loss costs to enhance the overall cost efficiency of the distribution system. The second objective function, introduced in (2), aims to improve the voltage profile across the distribution network. The proposed VCI metric goes beyond the traditional approach of simply analyzing the average voltage across all buses and time periods. Instead, it incorporates both the deviation of voltage values from the nominal voltage (V^n) and the bounds of the voltage distribution, adjusted to exclude outliers. Specifically, the upper bound, σ_1 , is defined as the minimum between the largest observation in the dataset and the 75th percentile plus 1.5 times the interquartile range. Similarly, the lower bound, σ_2 , is given by the maximum between the smallest observation in the dataset and the 25th percentile minus 1.5 times the interquartile range.

The numerator in (3) is composed of two main components. The first term, $(|\sigma_1 - V^n| + |\sigma_2 - V^n|)$, quantifies the deviation of σ_1 and σ_2 from the nominal voltage. This term is designed to minimize voltage variations across the system's buses and throughout the evaluated time horizon. By bringing σ_1 and σ_2 closer to the nominal voltage, the system's reliability is enhanced, maintaining voltage levels

within safe operating limits even under adverse conditions. The second term, $|V_{h,i} - V^n|$, represents the deviation of the voltage at bus i and time h from the nominal voltage, ensuring that the VCI metric accurately reflects the proximity of the system's overall voltage to the reference value. The combination of these terms makes the VCI sensitive to both average and boundary voltage deviations (excluding outliers), providing a more comprehensive and reliable evaluation of the system's performance.

Constraints (7) and (8) define the current and voltage limits of the network, respectively. The active and reactive power balances at each bus are expressed in (9) and (10). The voltage drop along branch ij is described in (11), while constraint (12) links the square of the current and voltage with the active and reactive power flows in branch ij . These power flow modeling equations for radial distribution systems are based on the configuration of the distribution system illustrated in Fig. 1.

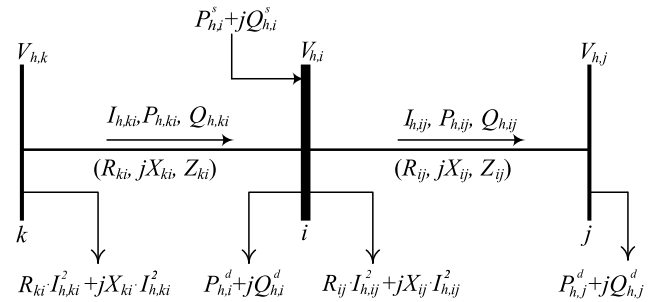


FIGURE 1. Distribution system representation [25].

The tap position limits of the OLTC transformer are defined in (13) and the substation bus voltage is expressed as a function of tap^{OLTC} in (14). Constraints (15)–(18) define the operating limits of the DERs following the IEEE 1547-2018 standard [26]. Additionally, constraint (19) ensures that the total investment cost remains within the available budget. Constraint (20) establishes that the number of closed switches is equal to the total number of buses minus one, which is necessary to maintain the radial topology of the network. However, this condition alone is not sufficient to guarantee a loop-free configuration. To fully ensure the radial structure, the approach proposed in [12] is employed, which systematically eliminates any loops and confirms the radiality of the system. Finally, the maximum number of switching operations is defined in (21), since frequent switching can not only compromise the system's stability but also shorten the lifespan of these devices, as discussed in [27], [28].

III. PROPOSED METHODOLOGY

In this article, the Multi-objective Cuckoo Search (MOCS) is adopted to solve the multi-objective optimization problem involving network reconfiguration, the allocation and sizing of capacitors, and the adjustment of the OLTC transformer at the substation. Nevertheless, the emphasis of this research is on the proposed methodology rather than the optimization

method itself. Fig. 2 illustrates the detailed steps of the proposed methodology for solving this multi-objective optimization problem. In Fig. 2, ite denotes the iteration counter of the MOCS and $\bar{ite}$ represents the maximum number of iterations.

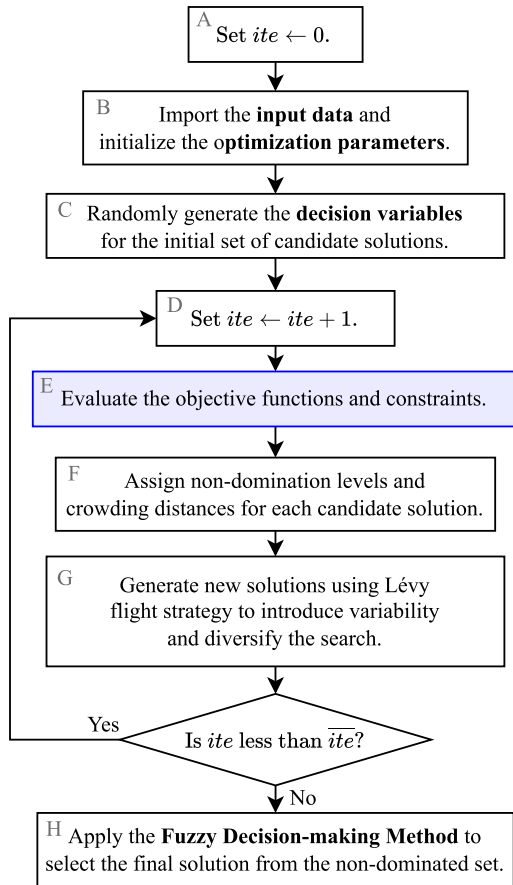


FIGURE 2. Flowchart of the proposed methodology.

The iteration counter of the MOCS is initially set to 0 in Block A. Thereafter, in Block B, the input data, which includes historical irradiance and load profiles, is imported. To ensure the methodology's reliability, the irradiance data should cover a sufficiently long time span to capture the seasonal variations throughout the year, reflecting the full range of DER power generation. Additionally, it is essential to account for various load profiles, such as those of residential and commercial customers, as these differ significantly and influence the system's performance. Regarding the MOCS, the parameters initialized are: number of nests, discovery probability for outlier solutions, and maximum number of iterations.

In Block C, the decision variables—comprising the capacitors locations, the status of sectionalizing and tie switches, the capacitors tap positions, and the OLTC transformer tap position—are randomly generated to form the initial solution. Subsequently, in Block D, the iteration counter is incremented by 1, moving the algorithm forward in

its optimization process. Furthermore, Block E handles the evaluation of the objective functions and constraints. The procedure for assessing these objective functions and constraints, considering customer-owned DERs, is detailed in Section III-A.

In Block F, the non-domination level and crowding distance are assigned to each candidate solution. Moreover, new solutions are generated using Lévy flights in Block G. After completing all $\bar{ite}$ iterations of the MOCS, the Fuzzy Decision-making Method is performed in Block H. This method involves defining membership functions, denoted as $\Phi_{f_a^n(x)}(\bar{x}_n)$, for each objective function $f_a^n(x)$ considering any available solution in the Pareto solution set $\bar{x}_n$. These membership functions quantify the satisfaction level of the decision maker with respect to the value of the objective function a for a given solution n , where 1 represents the highest degree of satisfaction and 0 indicates no satisfaction.

In this article, the linear membership function is adopted due to its simplicity and ability to provide a straightforward representation of the decision maker's preferences. The mathematical expression for the linear membership function is expressed in (22). In the Fuzzy Decision-making Method, the next step involves determining the level of desirability associated with each objective function $f_a(x)$, which is represented by $\tilde{\Phi}_{f_a(x)}$. Therefore, the compromise solution can be found using (23).

$$\Phi_{f_a^n(x)}(\bar{x}_n) = \begin{cases} 0 & \forall \{f_a^n(x) > f_a^{\max}(x)\}, \\ \frac{f_a^{\max}(x) - f_a^n(x)}{f_a^{\max}(x) - f_a^{\min}(x)} & \forall \{f_a^{\min}(x) \leq f_a^n(x) \leq f_a^{\max}(x)\}, \\ 1 & \forall \{f_a^n(x) < f_a^{\min}(x)\}, \end{cases} \quad (22)$$

$$\min_{n \in \Omega_n} \left\{ \max_{a \in \Omega_a} \{ |\tilde{\Phi}_{f_a(x)} - \Phi_{f_a^n(x)}(x_n)| \} \right\}, \quad (23)$$

where Ω_a and Ω_n represent, respectively, the sets of objective functions and available solutions; and $f_a^{\max}(x)$ and $f_a^{\min}(x)$ are, respectively, the maximum and minimum values of the objective function $f_a^n(x)$.

A. OBJECTIVE FUNCTIONS AND CONSTRAINTS ASSESSMENT

Fig. 3 provides a detailed representation of the steps involved in evaluating the objective functions and constraints, previously outlined in Block E of the flowchart in Fig. 2. In Fig. 3, t represents the time counter (in hours), g refers to the counter of scenarios generated by the Monte Carlo Method, and t^{int} indicates the multi-period time interval of the multi-period analysis (in hours). The maximum value for the time counter is $\bar{t}$, while the maximum number of scenarios generated by the Monte Carlo Method is $\bar{g}$.

The procedure depicted in Fig. 3 starts with Block E1, where the time and Monte Carlo scenario counters are

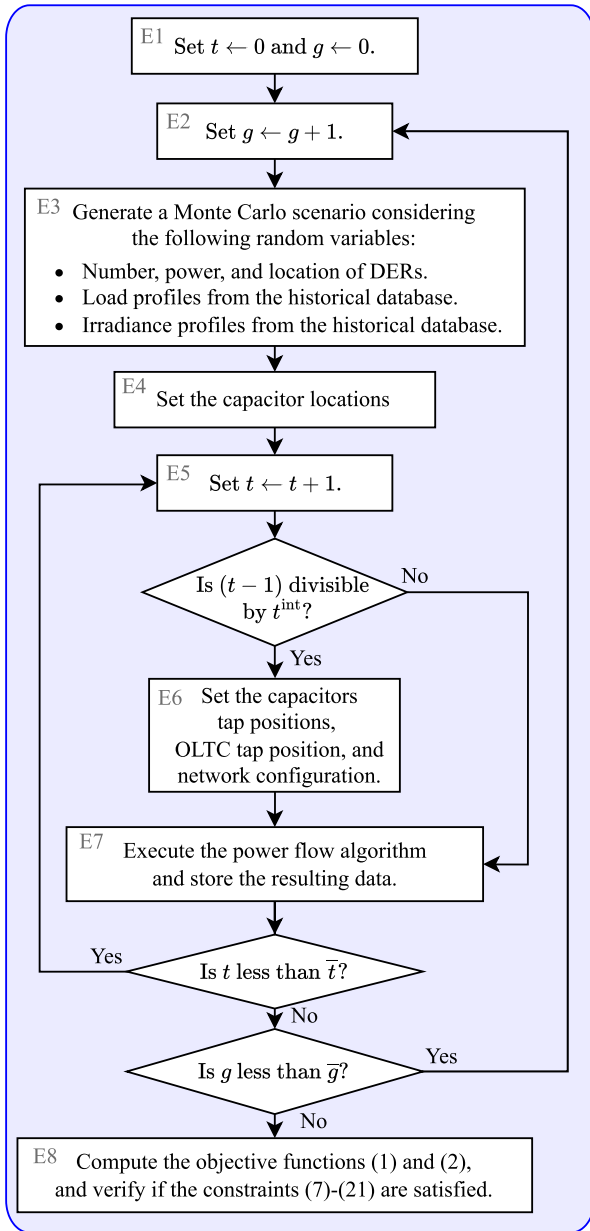


FIGURE 3. Flowchart illustrating the procedure for assessing the objective functions and verifying constraints satisfaction.

initialized to 0. In Block E2, the Monte Carlo scenario counter is subsequently increased by 1. In Block E3, a Monte Carlo scenario is randomly generated considering the number, location, and power of DERs, alongside load and irradiance profiles from the historical database. Regarding the DERs, the number of generators ranges from 0 to the number of customers of the system, their sizes range from 0% to 100% of the respective customers' power consumption, and their locations are randomly distributed across the distribution system.

Then, considering the decision variables defined by the optimization algorithm, the capacitor locations are config-

ured in Block E4. The time counter is incremented in Block E5 and, for specific multi-period intervals, the capacitor tap positions, OLTC tap settings, and network configuration are updated in Block E6 according to the decision variables. After performing the power flow analysis for each time of the day and each Monte Carlo scenario, the objective functions (1) and (2) are computed in Block E8 under constraints (7)–(21), ensuring compliance with operational and physical limits of the distribution system.

IV. RESULTS AND DISCUSSION

This section presents an overview of the case study, detailing the distribution system, input data, and simulation parameters. It then evaluates the impacts of stochastic customer-owned DERs and discusses the optimization results obtained through the proposed approach.

A. CASE STUDY

The proposed approach integrates the allocation and sizing of capacitors, the operation of the OLTC transformer, and network reconfiguration. To assess the individual impact of each strategy on the objective functions, the following approaches are tested:

- 1) Capacitors allocation and sizing.
- 2) OLTC transformer operation.
- 3) Network reconfiguration.
- 4) Capacitors allocation and sizing combined with OLTC transformer operation.
- 5) Capacitors allocation and sizing combined with network reconfiguration.
- 6) Network reconfiguration combined with OLTC transformer operation.
- 7) Proposed approach: capacitors allocation and sizing, OLTC transformer operation, and network reconfiguration.

To evaluate and compare the performance of static and dynamic strategies, five configurations with different multi-period time intervals are analyzed: one static evaluation ($t^{int} = \bar{t}$) and four dynamic evaluations with multi-period time intervals of 8, 4, 2, and 1 h.

For the MOCS, simulations are executed with 100 nests, incorporating a 25% discovery probability and a maximum of 100 iterations. The voltage lower and upper limits are 0.88 and 1.10 pu, respectively, based on the IEEE 1547-2018 standard [26]; while the current limits are defined according to each line section. The OLTC transformer has a voltage step of 1.8% per tap and 19 operating positions (± 9 from the neutral position) with an investment cost of 7,000 US\$ [29]. The capacitors available for allocation feature a nominal reactive power of 175 kVar, divided into 7 equally switchable taps, with an investment cost of 30 US\$/kVar [30]. The total available investment budget is 35,000 US\$. Additionally, the energy cost is 0.13 US\$/kWh [30]. The maximum number of switching operations, represented by Δs^{dyn} , is 6.

The maximum number of random scenarios generated by the Monte Carlo Method is 100. For the Fuzzy Decision-making Method, the desirability levels $\tilde{\Phi}_{f_1(x)}$ and $\tilde{\Phi}_{f_2(x)}$ are set to 0.5 and 1.0, respectively. Additionally, the proposed methodology is evaluated using the 69-bus test system, shown in Fig. 4, which has 68 sectionalizing switches and 5 tie switches. The choice of this particular system is motivated by its size, which allows for a clearer presentation and discussion of the obtained results. Additional parameters of this system are available in [31].

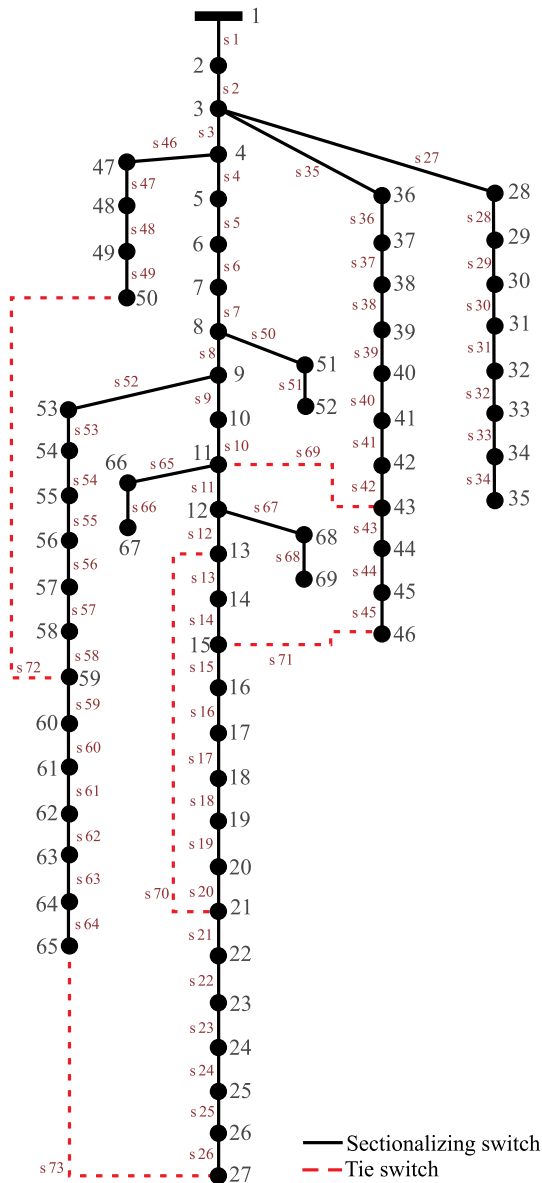


FIGURE 4. Representation of the 69-bus test system.

Fig. 5 shows the load profile and solar irradiance data collected by the National Institute of Meteorology [32] and Energisa [33], respectively, over a three-year period (2018–2020) in Campina Grande, Brazil. The figure also includes the average values for each multi-period time interval,

where the averages for the 24-hour, 8-hour, 4-hour, and 2-hour intervals are depicted in black, red, blue, and green, respectively.

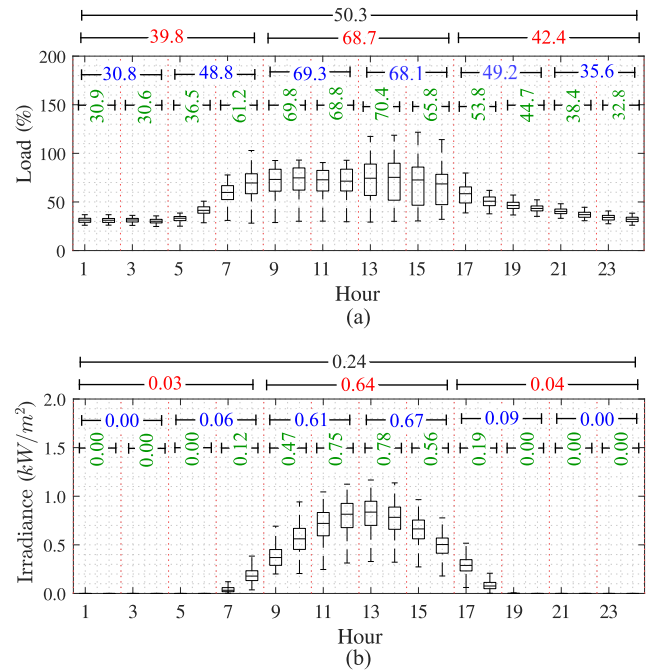


FIGURE 5. (a) Daily load profile and (b) solar irradiance data collected over three years (2018–2020) in Campina Grande, Brazil, with the average values calculated for each multi-period time interval.

Moreover, the Pearson correlation coefficient is used in this study to evaluate the relationship between the decision variable results and the characteristics of the distribution network. The Pearson correlation coefficient is a widely used measure of linear association, which quantifies the strength and direction of a linear relationship between two variables. This dimensionless coefficient is based on the covariance of the variables and is scaled to range from -1 to $+1$. A coefficient of $+1$ indicates a perfect positive linear correlation, -1 indicates a perfect negative linear correlation, and 0 indicates no linear correlation [34].

B. IMPACTS OF STOCHASTIC CUSTOMER-OWNED DERS

Figs. 6 and 7 present the voltage profile and power losses of the original system with and without stochastic customer-owned DERs, respectively. These figures are provided to illustrate the impacts of DERs integration on the system's performance. In Figs. 6a and 7a, each point represents a voltage value measured at each bus of the system for every random scenario generated using the Monte Carlo Method. Similarly, in Figs. 6b and 7b, each point corresponds to a power loss value measured at each line of the system under the same set of random scenarios. It is important to note that the points in the graphs are plotted with 2% opacity to allow for a clear visualization of the density and distribution of data points in each region. This approach effectively highlights areas where voltage or power loss values are more

concentrated, facilitating an in-depth understanding of the variability introduced by the stochastic nature of DERs.

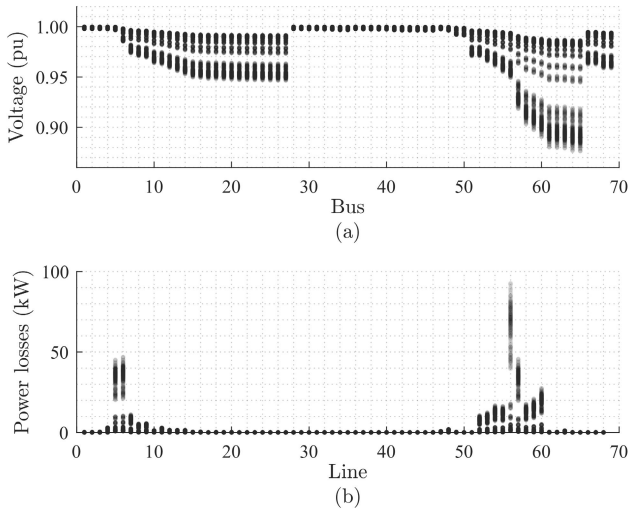


FIGURE 6. (a) Voltage profile at each bus and (b) power losses at each line considering the original system without stochastic customer-owned DERs.

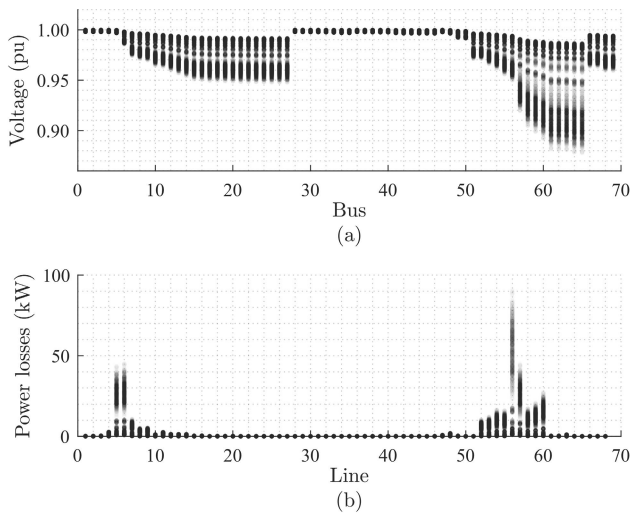


FIGURE 7. (a) Voltage profile at each bus and (b) power losses at each line considering the original system with stochastic customer-owned DERs.

When comparing the voltage profiles and power losses in Figs. 6 and 7, it is possible to notice that the minimum voltage values and maximum power losses change only slightly with the integration of stochastic DERs. This subtle variation occurs because DERs are randomly distributed in terms of quantity, location, and capacity. As a result, certain scenarios may arise where few or no DERs are installed, leaving the system's performance in these cases similar to the original configuration without DERs.

Despite that, an improvement can be observed in other regions of the graphs. Many points in the voltage profile (Figs. 6a and 7a) shift upward, driven by scenarios where higher DER capacities are installed, leading to an increase in

the average voltage across the system. Similarly, reductions in power losses are evident in several points of the loss distribution (Figs. 6b and 7b), corresponding to scenarios where DERs effectively supply local demand or inject power into the grid. This results in a lower average for power losses after the DER integration.

Furthermore, the distribution of points in both voltage and power loss graphs becomes more dispersed when considering DERs. This increased spread reflects the expanded operational possibilities introduced by DERs, which adds variability to the system's performance. Consequently, this variability highlights the increased complexity of the system's analysis, requiring a more thorough consideration of uncertainties in operational conditions.

C. OPTIMIZATION RESULTS

Figs. 8a–8c illustrate the Pareto front approximation obtained by executing each strategy individually, including capacitors allocation and sizing, OLTC transformer operation, and network reconfiguration, respectively. To enhance visualization and address the overlapping of multiple points, each Pareto front approximation was shifted along the third axis of the graph. This adjustment was also applied to the graphs in Figs. 9 and 10. It is important to note that the objective function values f_1 and f_2 for the original system considering customer-owned DERs are 319.92 US\$ and 0.0287 pu, respectively.

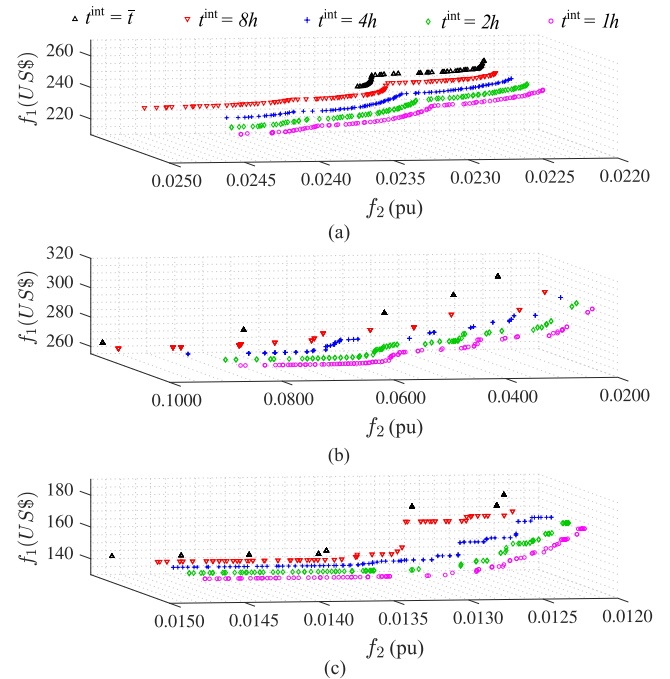


FIGURE 8. Pareto front approximation considering (a) capacitors allocation and sizing; (b) OLTC transformer operation; and (c) network reconfiguration.

Regarding the allocation of capacitors, shown in Fig. 8a, capacitors are primarily placed at buses with high load demand and those located farther from the substation. This

allocation pattern is supported by a positive correlation (Pearson coefficient of 0.74) between the installed capacitor power and the total load at each bus of the distribution network. Additionally, a positive correlation (Pearson coefficient of 0.37) is also observed between the installed capacitor power and the total impedance separating each bus from the substation.

Furthermore, in the multi-period configurations, capacitors are adjusted to higher tap positions during periods of increased load demand, when reactive power support is essential for maintaining the system's voltage profile. In this context, a positive correlation (Pearson coefficient of 0.85) is observed between the capacitor tap position and the total system load. Similarly, for the OLTC transformer operation, shown in Fig. 8b, a strong positive correlation (Pearson coefficient of 0.91) exists between the OLTC tap position and the overall system load demand.

During network reconfiguration, it is observed that the multi-objective optimization algorithm defines topologies that strategically reduce the total distance between buses with heavier loads and the substation, while increasing the distance for buses with lighter loads. These changes are measured relative to the original configuration, in which switches s_{69} , s_{70} , s_{71} , s_{72} , and s_{73} , presented in Fig. 4, are open. A negative correlation (Pearson coefficient of -0.36) was calculated between each bus's load and its distance variation to the substation, where a negative distance variation indicates a reduction and a positive variation represents an increase compared to the original system's configuration. By minimizing the distance between heavier loads and the substation, the power losses are decreased, since the current travels shorter paths to reach high-demand buses, consequently improving the system's voltage profile.

The results presented in Fig. 8 highlight that subdividing the multi-period time intervals increases the diversity of solutions and improves the quality of the results by expanding the search space. However, this improvement diminishes progressively as the number of intervals increases. This trend aligns with the load and irradiance patterns shown in Fig. 5, where significant variations in average values are observed when comparing the entire day (1–24 h) to three equally divided periods (1–8 h, 9–16 h, and 17–24 h). In contrast, further subdividing these three intervals does not substantially affect the average load and irradiance values. In other words, the variation in these averages decreases as the number of intervals grows, meaning that additional subdivisions primarily serve to enhance solution diversity by further expanding the search space of the optimization problem.

Figs. 9a–9c show the Pareto front approximation considering, respectively, the OLTC transformer operation combined with capacitors allocation and sizing; network reconfiguration combined with capacitors allocation and sizing; and network reconfiguration combined with OLTC transformer operation. Furthermore, Fig. 10 illustrates the Pareto front approximation considering the combination of

network reconfiguration, OLTC transformer operation, and capacitors allocation and sizing.

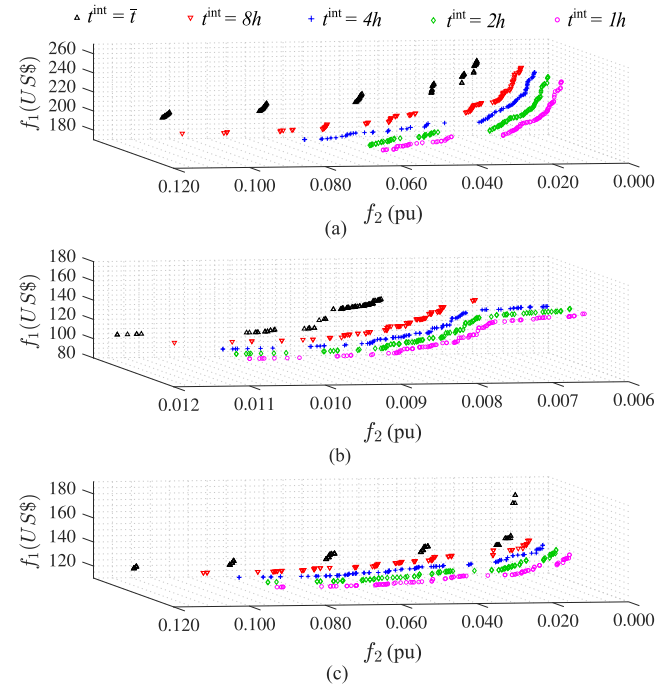


FIGURE 9. Pareto front approximation considering (a) OLTC transformer operation combined with capacitors allocation and sizing; (b) network reconfiguration combined with capacitors allocation and sizing; and (c) network reconfiguration combined with OLTC transformer operation.

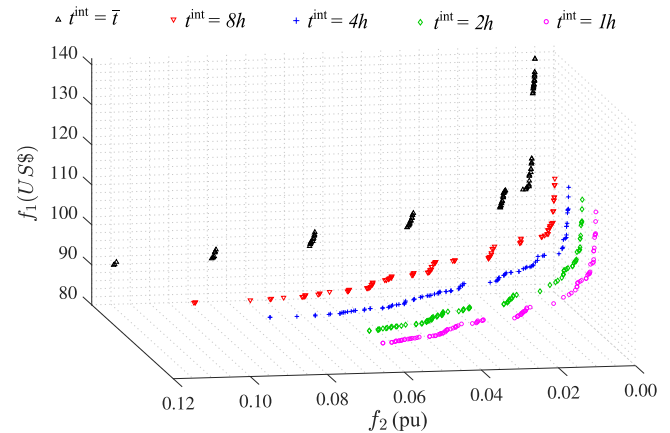


FIGURE 10. Pareto front approximation considering the combination of network reconfiguration, OLTC transformer operation, and capacitors allocation and sizing.

As expected, integrating the strategies into a multi-objective optimization problem significantly expands the search space, resulting in higher-quality solutions. From Figs. 8b, 9a, 9c, and 10, it can be observed that the operation of the OLTC transformer has a substantial impact on the objective function f_2 . In other words, while adjusting the OLTC transformer tap can improve the system's voltage profile, it can also degrade it, potentially making the system more unstable and less

reliable. Therefore, properly incorporating this operational aspect into system planning is essential to ensure both stability and reliability.

Conversely, as illustrated in Figs. 8a, 9a, 9b, and 10, the allocation of capacitors has a limited impact on the objective function f_2 . However, it is highly effective in reducing costs associated with power losses. Consequently, the optimized allocation and sizing of capacitors in the distribution system enhance the overall efficiency while providing modest improvements to the voltage profile. With respect to network reconfiguration, as shown in Figs. 8c, 9b, 9c, and 10, this strategy has the potential to significantly reduce both objective functions. It demonstrates notable improvements in the voltage profile of the system, as well as reductions in costs associated with power losses.

Similarly to what is depicted in Fig. 8, the results shown in Figs. 9 and 10 reveal diminishing returns as the multi-period time intervals t^{int} are further subdivided. Despite that, the most significant reductions are observed when the multi-period time interval is set to 1 h. For this reason, the Pareto front approximations of the seven evaluated approaches are presented in Fig. 11, considering a multi-period time interval of 1 h.

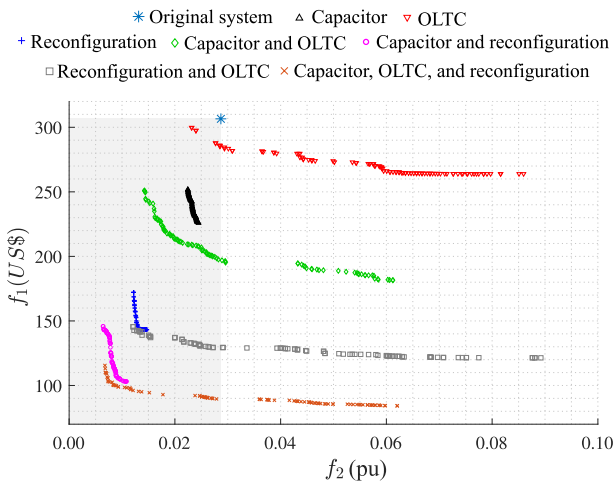


FIGURE 11. Pareto front approximation considering the evaluated approaches and $t^{\text{int}} = 1$ h.

Fig. 11 demonstrates that all the identified solutions achieve lower costs associated with power losses compared to the original system configuration. Additionally, when analyzing the three individual strategies, it is evident that all solutions obtained through the operation of the OLTC transformer are dominated by at least one solution derived from the allocation of capacitors. Similarly, all solutions resulting from the allocation of capacitors are dominated by at least one solution obtained through the dynamic network reconfiguration. Therefore, in the case study analyzed, dynamic network reconfiguration proves to be the most effective strategy for minimizing power loss costs and enhancing the system's voltage profile, while the OLTC transformer operation is the least effective.

A similar observation can be made for the combined approaches. The combination of capacitors allocation and dynamic network reconfiguration stands out as the most effective strategy for minimizing both f_1 and f_2 , whereas the combination of capacitors allocation with OLTC transformer operation is the least effective. Moreover, the integration of all three strategies—capacitors allocation, OLTC transformer operation, and dynamic network reconfiguration—yields the best overall results.

Subsequently, the Fuzzy Decision-making Method is employed to determine the final solution for each approach, considering a multi-period time interval of 1 h. Fig. 12 presents a histogram of the voltage profile of the distribution system, where the frequency values are normalized such that the highest frequency value reaches 1. Additionally, a boxplot is overlaid to illustrate the interquartile range as well as the highest and lowest data points, excluding any outliers, of the voltage profile dataset. It is important to mention that the results presented in Figs. 12 and 13 account for the integration of customer-owned DERs.

In Fig. 12a, which corresponds to the original system, the voltage profile shows a high concentration of values near the lower acceptable limit, reflecting improper voltage regulation and a lack of reliability in the system. This serves as a baseline for evaluating the impact of the proposed optimization strategies. It is important to mention that the wide variation in voltage values arises from uncertainties related to load, DER generation, and the quantity, sizes, and locations of customer-owned DERs. In Fig. 12b, where only capacitors allocation and sizing are considered, there is a shift of the voltage values closer to the nominal voltage, resulting in an improved profile. However, this approach still exhibits high variability, suggesting limited effectiveness when employed as a standalone solution.

Fig. 12c, representing OLTC transformer operation, shows a slight improvement in the voltage profile compared to the original system. The adjustment of the OLTC transformer results in voltage values shifting closer to the upper acceptable limit. While this indicates better regulation compared to the baseline, the method remains insufficient when applied alone, given the proximity of some voltage values to the acceptable bounds. Conversely, Fig. 12d, which reflects the results of network reconfiguration, achieves substantial improvements, with voltage values more concentrated near the nominal voltage. This demonstrates the superior impact of reconfiguration compared to the isolated application of OLTC operation or capacitors allocation.

In Fig. 12e, the combination of OLTC operation and capacitors allocation achieves better voltage regulation than their individual applications. However, this approach is still outperformed by the results obtained through network reconfiguration. Fig. 12f, which integrates network reconfiguration with capacitors allocation, demonstrates a significant improvement, presenting a tighter distribution of voltage values near the nominal voltage and minimizing

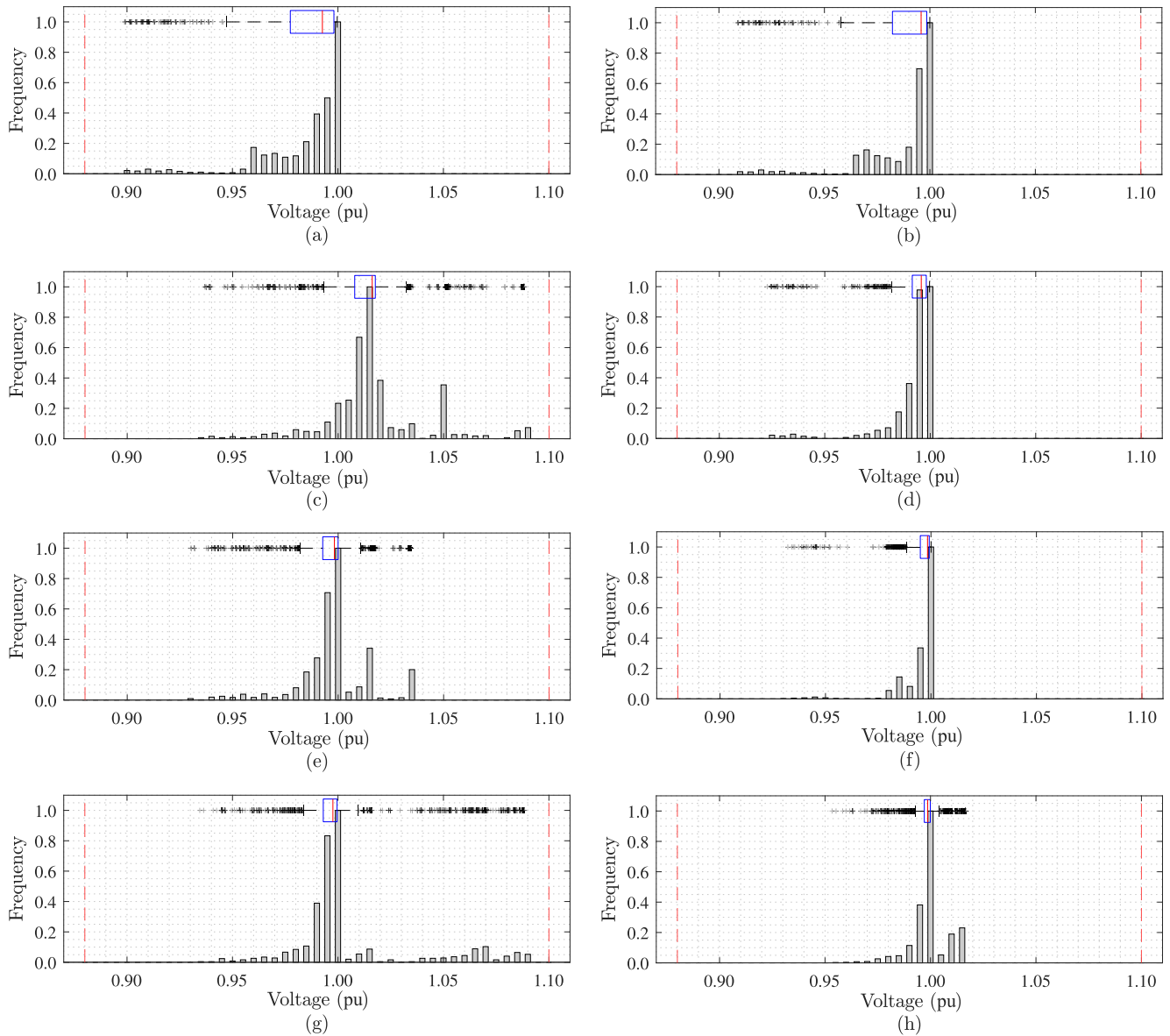


FIGURE 12. Voltage profile considering the (a) original system; (b) capacitors allocation and sizing; (c) OLTC transformer operation; (d) network reconfiguration; (e) OLTC transformer operation combined with capacitors allocation and sizing; (f) network reconfiguration combined with capacitors allocation and sizing; (g) network reconfiguration combined with OLTC transformer operation; and (h) combination of network reconfiguration, OLTC transformer operation, and capacitors allocation and sizing.

extreme deviations. On the other hand, Fig. 12g, which combines network reconfiguration with OLTC operation, also enhances the voltage profile but falls slightly short compared to Fig. 12f. This comparison highlights that the inclusion of capacitors has a greater impact on improving voltage regulation than OLTC operation when paired with network reconfiguration. Finally, Fig. 12h, which integrates all three strategies, achieves the most optimized voltage profile. However, while this comprehensive approach produces the best results, the improvement over Fig. 12f is relatively marginal, suggesting diminishing returns from adding OLTC operation to the combination of reconfiguration and capacitors, when considering only the VCI.

Fig. 13 shows the average power losses for each line and hour, considering the original system and the seven approaches evaluated in this article. In Fig. 13a, which represents the power losses in the original system, the losses are concentrated along specific lines with higher impedance and close to heavier loads, with notable peaks in time period when the load is higher, as shown in Fig. 5a. These high peaks highlight inefficiencies in the system under the baseline configuration, with no interventions applied. Fig. 13b, showing the effect of the allocation and sizing of capacitors, illustrates a reduction in power losses compared to the original system. While the peaks remain in the same regions, their magnitudes are reduced, indicating an

improvement in the system's efficiency due to reactive power compensation. In Fig. 13c, where OLTC transformer operation is implemented, a slight improvement in power loss reduction is observed. However, the effect is less pronounced than in Fig. 13b, as the OLTC operation alone is not as effective in reducing losses across the network. The distribution of losses remains similar to the original system, although the magnitude of some peaks is diminished.

Fig. 13d highlights the performance of network reconfiguration, where power losses are minimized compared to Figs. 13b and 13c. Unlike the original system or the other individual strategies, the losses become more concentrated along a specific line rather than being distributed across multiple lines. This shift indicates a reallocation of power flows in the network. Despite the overall minimization of losses, concentrating them on fewer lines may not be ideal in the long term, as it could necessitate network expansion to accommodate increased load demands. Over time, this could result in higher infrastructure costs and operational challenges, making a more balanced distribution of power losses a potentially more sustainable approach.

On the other hand, Fig. 13e combines OLTC operation with capacitors allocation and sizing, yielding a more prominent reduction in power losses compared to their individual effects. The peaks are further suppressed, but they still remain in the same lines as the original system. Figs. 13f and 13g, which combine network reconfiguration with capacitors allocation and OLTC operation, respectively, present similar behaviors in reducing power losses across the system. Both strategies achieve notable improvements, with the lines experiencing higher losses now concentrated in different areas compared to the original system, due to the reconfiguration process. However, the combination with capacitors allocation (Fig. 13f) results in slightly more effective loss reduction than the combination with OLTC operation (Fig. 13g). Finally, Fig. 13h, representing the combined application of all three strategies, achieves the lowest power losses among all configurations, indicating that the integration of all methods offers significant benefits. Additionally, this approach significantly reduces the peak of the losses, which may contribute to greater long-term efficiency by potentially delaying or minimizing the need for future network expansion investments.

Table 2 presents the objective function values for each approach evaluated in this study, considering $t^{\text{int}} = 1$ h. The results in Table 2 support the insights derived from the analyses of Figs. 12 and 13, emphasizing the effectiveness of network reconfiguration, both individually and when combined with other strategies. Notably, network reconfiguration alone achieves a 49.99% reduction in f_1 compared to the original system, highlighting its significant impact on reducing power loss costs. When combined with capacitors allocation, f_1 is further reduced by 14.56%, while its combination with OLTC transformer operation results in a further reduction of 15.11%. The comprehensive strategy that combines capacitors allocation, OLTC operation, and

network reconfiguration achieves the lowest f_1 , representing a 37.13% reduction compared to network reconfiguration alone and a 68.56% reduction compared to the original system.

TABLE 2. Values of the objective functions for each approach considering $t^{\text{int}} = 1$ h.

Approach	f_1 (US\$)	f_2 (pu)
Original system	319.92	0.029
Capacitors allocation and sizing	244.61	0.023
OLTC transformer operation	284.56	0.029
Network reconfiguration	159.97	0.012
Capacitors allocation and sizing combined with OLTC transformer operation	220.89	0.018
Capacitors allocation and sizing combined with network reconfiguration	136.67	0.008
Network reconfiguration combined with OLTC transformer operation	135.79	0.021
Capacitors allocation and sizing, OLTC transformer operation, and network reconfiguration	100.57	0.008

In terms of f_2 , network reconfiguration results in a 58.62% reduction compared to the original system, while the comprehensive approach achieves a total reduction of 72.41%. It can be observed that, although the OLTC transformer operation directly influences the voltage profile, this strategy alone is not effective in minimizing f_2 , as it produced a VCI value nearly identical to that of the original system. This is because the strategy only adjusts the voltage at the substation, whereas the critical issue lies in the voltage variation throughout the network. As the impedance between the substation and the bus increases, the voltage decreases, approaching the lower acceptable limit.

Furthermore, it is essential to assess the practical implications for utility operators when implementing the proposed methodology. In the approach where capacitors are allocated and sized, an OLTC transformer is installed and operated, and the system is reconfigured, the total investment cost is 33,250 US\$, which remains within the available investment budget. Conversely, in the original system, since no expansion is carried out, there are no investment costs. However, the original system incurs an active power loss cost of 319.92 US\$ per day, whereas the proposed approach, incorporating capacitors, the OLTC transformer, and system reconfiguration, results in a significantly lower active power loss cost of 100.57 US\$ per day. Based on these values, the payback period for the investment in network expansion is approximately 152 days, given the substantial reduction in active power loss costs.

Beyond the financial aspects, feasibility must also be considered when deploying these strategies in real-world distribution networks. The integration of OLTC transformers and capacitors requires careful planning to ensure compatibility with existing infrastructure and compliance with regulatory standards. Additionally, system reconfiguration may introduce operational challenges, such as switching constraints and reliability concerns, which must be addressed through proper coordination and monitoring. Despite these challenges, the relatively short payback period and long-term

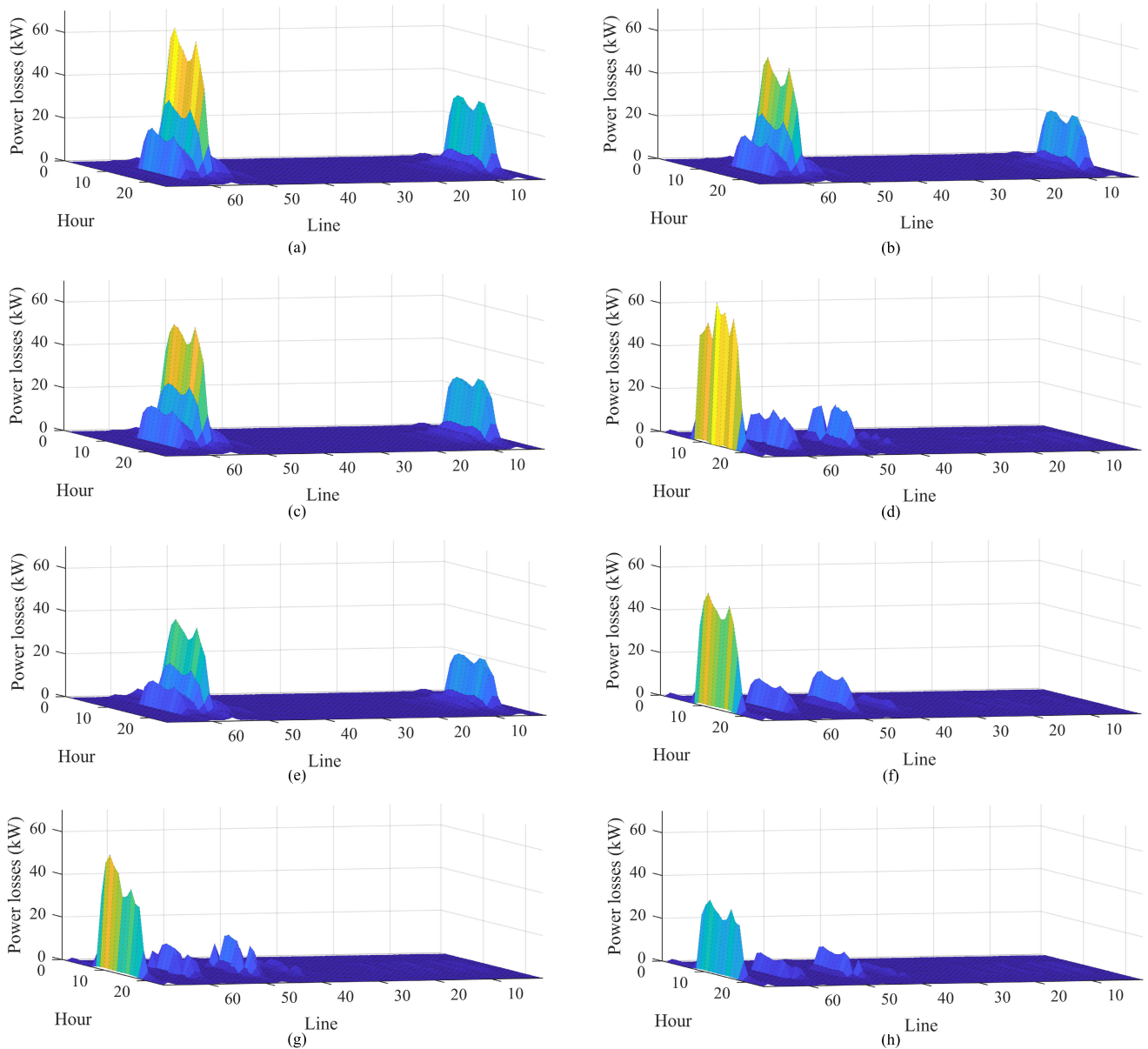


FIGURE 13. Power losses considering the (a) original system; (b) capacitors allocation and sizing; (c) OLTC transformer operation; (d) network reconfiguration; (e) OLTC transformer operation combined with capacitors allocation and sizing; (f) network reconfiguration combined with capacitors allocation and sizing; (g) network reconfiguration combined with OLTC transformer operation; and (h) combination of network reconfiguration, OLTC transformer operation, and capacitors allocation and sizing.

cost savings indicate that the proposed approach is a viable and economically attractive solution for improving distribution system performance. Moreover, these strategies are already widely adopted in practical distribution networks, further reinforcing their feasibility.

V. CONCLUSION

In this article, a multi-objective planning approach was proposed to optimize the performance of the distribution system by simultaneously addressing dynamic network reconfiguration, capacitors allocation, and OLTC transformer adjustment. The approach incorporated stochastic

customer-owned DERs and introduced a novel VCI to enhance voltage regulation while minimizing power loss costs. The results highlighted the effectiveness of dynamic network reconfiguration, both as a standalone strategy and when combined with other methods, particularly capacitors allocation and OLTC transformer adjustment.

Based on the findings, the significant expansion of the search space achieved by integrating the strategies resulted in higher-quality solutions. In this context, the integrated approach, combining all three strategies, yielded the best results, reducing power loss costs by 68.56% and significantly improving voltage profile, with a 72.41% reduction in

the VCI compared to the original system. It is also noteworthy that this approach resulted in a reduction in the peaks of power losses, contributing to improved long-term efficiency by potentially postponing or reducing the need for future network expansion investments.

Finally, the incorporation of stochastic modeling ensured reliable solutions under varying conditions, while the use of the MOCS and Fuzzy Decision-making Method provided an effective mechanism for balancing competing objectives. Compared to existing methodologies, this study demonstrated the advantages of an integrated and dynamic approach, offering a practical framework for optimizing distribution networks with high penetration of stochastic customer-owned DERs. This work underscores the importance of coordinated planning in modern power systems, presenting an effective solution for enhancing the efficiency and reliability in distribution system operations.

Despite the advances made in this article, the study considered only DERs based on photovoltaic systems and was limited to the 69-bus test system. Therefore, future research should evaluate the proposed methodology on larger-scale distribution systems and in scenarios with highly dynamic DERs to further assess its effectiveness and scalability. Additionally, incorporating machine learning or artificial intelligence-based methods could further improve some aspects of the proposed approach, including the modeling of uncertainties associated with DER integration and operation, optimizing decision-making processes, and enabling adaptive strategies that better respond to dynamic changes in distribution networks.

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energy resources, power system protection, artificial intelligence, meta-heuristic optimization methods, multi-objective optimization, and modeling and analysis of inverter-based generators.

RENATO S. F. FERRAZ received the B.Sc. degree in electrical engineering from the State University of Santa Cruz, Ilhéus, Bahia, Brazil, in 2018, the M.Sc. degree in electrical engineering from the Federal University of Espírito Santo, in 2020, and the Ph.D. degree in electrical engineering, with a focus on power processing and electrical systems from the Federal University of Espírito Santo, Vitória, Espírito Santo, Brazil, in 2025.

His current research interests include distributed energy resources, power system protection, artificial intelligence, meta-heuristic optimization methods, multi-objective optimization, and modeling and analysis of inverter-based generators.



He is currently a Professor with the Center for Open and Distance Education, Federal University of Piauí. During his undergraduate studies, he spent time at Arizona State University, USA, and conducted research with Illinois Institute of Technology in the field of computer engineering, in 2015. His current research interests include artificial intelligence, distributed energy resources, electric vehicles, metaheuristic optimization methods, multi-objective optimization, and time-series forecasting.

RAFAEL S. F. FERRAZ received the B.Sc. degree in electrical engineering from the State University of Santa Cruz, Ilhéus, Bahia, Brazil, in 2018, the M.Sc. degree in electrical engineering from the Federal University of Espírito Santo, in 2020, and the Ph.D. degree in electrical engineering, with a focus on power processing and electrical systems from the Federal University of Espírito Santo, Vitória, Espírito Santo, Brazil, in 2025.

He is currently a Professor with the Center for Open and Distance Education, Federal University of Piauí. During his undergraduate studies, he spent time at Arizona State University, USA, and conducted research with Illinois Institute of Technology in the field of computer engineering, in 2015. His current research interests include artificial intelligence, distributed energy resources, electric vehicles, metaheuristic optimization methods, multi-objective optimization, and time-series forecasting.



resources, multi-objective optimization, fuzzy logic, and decision-making methods.

VANIO F. S. JÚNIOR received the B.Sc. degree in electrical engineering from the State University of Santa Cruz, Ilhéus, Bahia, Brazil, in 2021, and the M.Sc. degree in electrical engineering from the Federal University of Espírito Santo, in 2024. He is currently pursuing the Ph.D. degree in computational modeling in science and technology, with a focus on electrical systems with the State University of Santa Cruz. His current research interests include distributed energy



control of electrical distribution and transmission systems, acting on the following subjects: transmission expansion planning, distributed generation (wind turbines, photovoltaic systems, small hydro plants, and biomass-based plants), electric power markets, electrical machines, mathematical optimization methods applied for solving linear and non-linear problems, multi-objective optimization, and statistical methods for temporal series forecasting.

AUGUSTO C. RUEDA-MEDINA received the bachelor's and M.Sc. degrees in electrical engineering from the Universidad Tecnológica de Pereira, in 2005 and 2008, respectively, and the Ph.D. degree in electrical engineering from UNESP, in 2012. He is currently a Professor with the Electrical Engineering Department, Federal University of Espírito Santo. He was a Postdoctoral Researcher with São Paulo State University. He has experience in analysis, operation, and

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